



Exploration of Deep Learning Methods for Synthetic T2-Weighted Pelvic MRI Generation from CT Scans: A Technical Feasibility Study

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Abstract

Synthesizing T2-weighted MRI from CT scans presents a challenging ill-posed problem that remains underexplored in abdominopelvic imaging. We aim to develop and compare deep learning algorithms for generating synthetic T2-weighted MRI from pelvic CT, systematically evaluating architecture and training strategies for feasibility and performance. The framework adopts a conditional Generative Adversarial Network (GAN) approach. Three state-of-the-art models [efficient Self-Attention UNet (ESAUNet), Residual Vision Transformer (ResViT), and Cascaded Gaze] were utilized as generators. A combined loss function (L1, VGG19 perceptual, adversarial) was employed to optimize fidelity. A multicenter cohort ($n=90$) including the SynthRad2023 ($n=60$) and in-house ($n=30$) datasets was used for model training. Five-fold cross-validation was used, and independent testing was performed on the Gold Atlas male cohort ($n=19$). Blinded qualitative assessment by two radiologists was conducted. Rigorous qualitative and quantitative assessments were performed. On the held-out cohort, ESAUNet achieved the highest average performance across all metrics: PSNR 22.21 dB, SSIM 0.748, and MAE 0.044. This surpassed the performance of the ResViT model (PSNR 21.18 dB, SSIM 0.727) and CascadedGazeNet (PSNR 21.41 dB, SSIM 0.703). Radiologists found no significant difference in overall pelvic image quality ($p=0.294$) or rectal wall delineation ($p=0.358$) between synthetic T2-weighted MRI and true MRI, despite lower ratings for fine details. Inter- and intra-observer agreement was robust. Conditional GAN architectures demonstrate technical feasibility for CT-to-T2-weighted pelvic MRI translation. ESAUNet's efficiency and robust performance highlight its potential as a key enabler for MRI-equivalent imaging in resource-limited settings.

Keywords Deep learning · Generative models · MRI · Pelvis · Translation

Peeyush Kumar Singh and Inam Ul Haq Gulzar contributed equally to this study.

Summary Statement Statement. An efficient deep learning model (ESAU-Net) can successfully synthesize T2-weighted pelvic MRI from CT scans, producing images with potential utility in computational applications and general anatomical review, thereby addressing a key gap in cross-modality image translation.

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Introduction

T2-weighted magnetic resonance imaging (T2w MRI) is the gold standard for pelvic cancer staging and radiotherapy planning [1]. Its unparalleled soft-tissue contrast enables precise delineation of tumors and organs at risk, directly impacting treatment accuracy and outcomes in prostate and rectal cancer. For these indications, T2w MRI is essential for visualizing tumor extent, extracapsular spread, and involvement of the rectal wall and sphincter, guiding contouring and dose planning for radiotherapy [1]. Yet, MRI is not universally accessible: high costs, scanner availability, and workflow complexities pose significant barriers, especially in non-tertiary and resource-limited settings.

The ability to generate synthetic MRI from CT, effectively unlocking MRI-like soft-tissue information from

CT scans, has transformative potential [2]. It would enable broader anatomical review, support computational research, and augment workflows by providing surrogate MRI data when actual scans are unavailable [3]. However, this task is technically formidable [4–6]. Unlike generating synthetic CT from MRI, the inverse CT-to-MRI transformation is an ill-posed problem [6]. CT images encode tissue density in Hounsfield units, which lack the biophysical basis for MRI's signal contrast. In the pelvis, this challenge is compounded by the anatomical complexity and the need to distinguish subtle structures with high fidelity. Prior deep learning approaches have focused on neuroimaging or synthetic CT; few have addressed pelvic synthetic MRI, and none have systematically benchmarked modern architectures on diverse, real-world datasets for T2w MRI synthesis [6–10].

This study introduces a conditional GAN framework leveraging state-of-the-art generator architectures generating synthetic T2-weighted MRI from pelvic CT. We compare three state-of-the-art architectures, ESAUNet [11], ResViT [12], and CascadedGazeNet [13], across a multi-center dataset.

Our main contributions are:

1. *Framework Novelty*: Integration of conditional GAN with advanced attention-based architectures and customized loss functions tailored for medical image translation.
2. *Dataset Diversity*: Utilization of multi-center datasets and preprocessing workflows to ensure generalizability.
3. *Comparison and Validation*: Systematic benchmarking of diverse architectures and radiologist assessments for clinical relevance.
4. *Impact*: Democratizing access to MRI-equivalent imaging via CT scans for pelvic oncology patients in resource-limited settings.

Materials and Methods

Study Design and Ethics

This study was conducted in accordance with the institutional review board guidelines. Ethics committee approval and written informed consent were obtained for in-house data.

Datasets

Three datasets were utilized:

For model training and validation:

- *SynthRad2023 dataset ($n=60$)*: The SynthRad2023 challenge was designed to benchmark and advance synthetic CT generation for MRI only or CBCT only radiotherapy planning in a multi-center, clinically representative set-

ting [14]. The dataset was collected from three Dutch university medical centers between 2018 and 2022 and comprised patients who underwent radiotherapy for pelvic (cervical, prostate, and rectal) tumors. For our task, we utilized a public dataset with paired CT and T2w MRI (3Tesla) from center C [14].

- *In-house dataset ($n=30$)*: This dataset comprised acute pancreatitis cases, including paired CT and standard 1.5 T T2w MRI [15]. Cases with extension of pancreatic fluid collection into the pelvis were excluded.

For held-out testing

Gold Atlas dataset ($n=19$): This dataset comprised independent test cohort of male patients with paired pelvic CT and MRI of prostate/rectal cancer, referred for curative radiotherapy [16].

The details of the dataset are provided in Table 1.

Data Preprocessing

Public Datasets

Both SynthRad 2023 and Gold Atlas dataset underwent standardized preprocessing prior to their public release. The preprocessing pipeline is described below:

SynthRad 2023 preprocessing pipeline included:

1. *File Conversion*: All DICOM images were converted to compressed NIfTI format (nii.gz).
2. *Resampling*: Images were resampled to a uniform voxel grid of $1 \times 1 \times 2.5 \text{ mm}^3$ to standardize spatial resolution.
3. *Image Registration*: Rigid registration was performed between paired scans using Elastix to ensure initial alignment.
4. *Anonymization*: All patient-identifying metadata was removed.
5. *Cropping*: Images were cropped to the bounding box of a patient outline mask (generated via ITK-based thresholding and hole-filling) with a 20-voxel margin to reduce file size and focus on relevant anatomy.

Pelvic Gold Atlas Dataset

Like the SynthRad, the Gold Atlas dataset had also undergone specialized preprocessing prior to its public release to handle its multi-rater delineations and to establish precise spatial correspondence between image pairs.

- *Consensus Delineation*: To create a ground-truth segmentation, a consensus was derived from multiple expert delineations using the Simultaneous Truth and Performance Level Estimation (STAPLE) algorithm, available via the Insight Segmentation and Registration Toolkit

Table 1 Dataset characteristics

Dataset	Patients	CT parameters	MRI parameters	Notes
SynthRad 2023 (Training, Center C only)	60	• Manufacturer: Philips-Siemens • Model: Brilliance BigBore • kVp: 120 • mA: 128–493 • Slice Thickness: 2–3 mm • Pixel Size: 0.98–1.17	• Manufacturer: Siemens • Model: MAGNETOM Avanto/Skyra/Vida • Field Strength: 3 T • Sequence: T2-weighted SPACE (FSE) • Acquisition: 3D • Flip Angle: 100–135° • Echo Numbers: 1 • TE/TR: 100–202/1500–2000 ms • Number of averages: 1–2 • Acquisition Matrix: 384,0,0,262	• Multi-center, radiotherapy planning • Standard clinical protocols • Real-world clinical heterogeneity
MRlencap (Training)	30	• Manufacturer: Siemens/Philips • Models: Definition Flash/iCT • kV: 120 • mA: 142–514 • Slice Thickness: 3 mm • Pixel Size: 0.6–1.4 mm²	• Manufacturer: Siemens • Model: MAGNETOM Aera • Field Strength: 1.5 T • Sequence: Axial T2-weighted HASTE • Acquisition: 2D • Flip Angle: 160° • TR/TE: 1000/98 ms • Number of averages: 1 • Slice Thickness: 4 mm • Acquisition Matrix: 320,262	• Acute pancreatitis • Clinical diagnostic scans
Gold Atlas (held-out test)	19	• Manufacturer: Siemens/Toshiba • Model: Somatom Definition, Aquilion, Emotion 6 • Slice Thickness: 2–3 mm • Pixel Size: ~1.0×1.0 mm² • Non-contrast	• Manufacturer: Siemens and GE • Discovery 750w • Sequence: FRFSE/TSE • Slice Thickness: 2.5 mm • Field Strength: 1.5 T • TR/TE: 6000–16,000/65–102	• Male patients • Prostate/rectal cancer • Fiducial markers present

FRFSE, fast recovery fast spin echo; *TSE*, turbo spin echo; *FFE*, fast field echo; *SPACE*, sampling perfection with application-optimized contrast using different flip-angle evolution; *FSE*, fast spin echo; *CTDIvol*, computed tomography dose index volume; *TR*, repetition time; *TE*, echo time

(ITK). This generated a probabilistic estimate of the true delineation, which was then refined by expert review to define a final consensus contour for each organ.

- **Deformable Image Registration:** A deformable registration was computed to map each CT image to its corresponding MR image using Elastix. The transformation was modelled with B-splines and optimized using mutual information as the similarity metric. As image information alone was insufficient for robust alignment, the registration was guided by anatomical contours: the STAPLE consensus mask was used for the MR bladder, and a model-based auto-segmentation from RayStation was used for the CT.

In-House Dataset

We adapted the SynthRad pipeline with a key modification. After initial conversion to NIFTI format, we utilized TotalSegmentator, a robust nnU-Net-based segmentation tool, to automatically generate masks for the sacrum [17]. These masks defined a consistent anatomical region of

interest, and both CT and MRI scans were subsequently cropped to the bounding box of these structures. This targeted cropping strategy effectively mitigated issues arising from asymmetric field of view and ensured that all scans were centered on the same pelvic anatomy. The remaining preprocessing steps, including resampling and registration, followed the same protocol as the SynthRad dataset.

Finally, for all datasets, the pre-processed 3D volumes were sliced into 2D axial images and saved in NumPy format (.npy), preparing them for input into our 2D synthesis models.

Deep Learning Framework Overview

The proposed framework employs a conditional GAN architecture, wherein CT slices serve as input to the generator network, tasked with synthesizing corresponding T2W MRI slices. A PatchGAN discriminator enforces realism by distinguishing synthetic from true MRI slices at a patch level (Fig. 1).

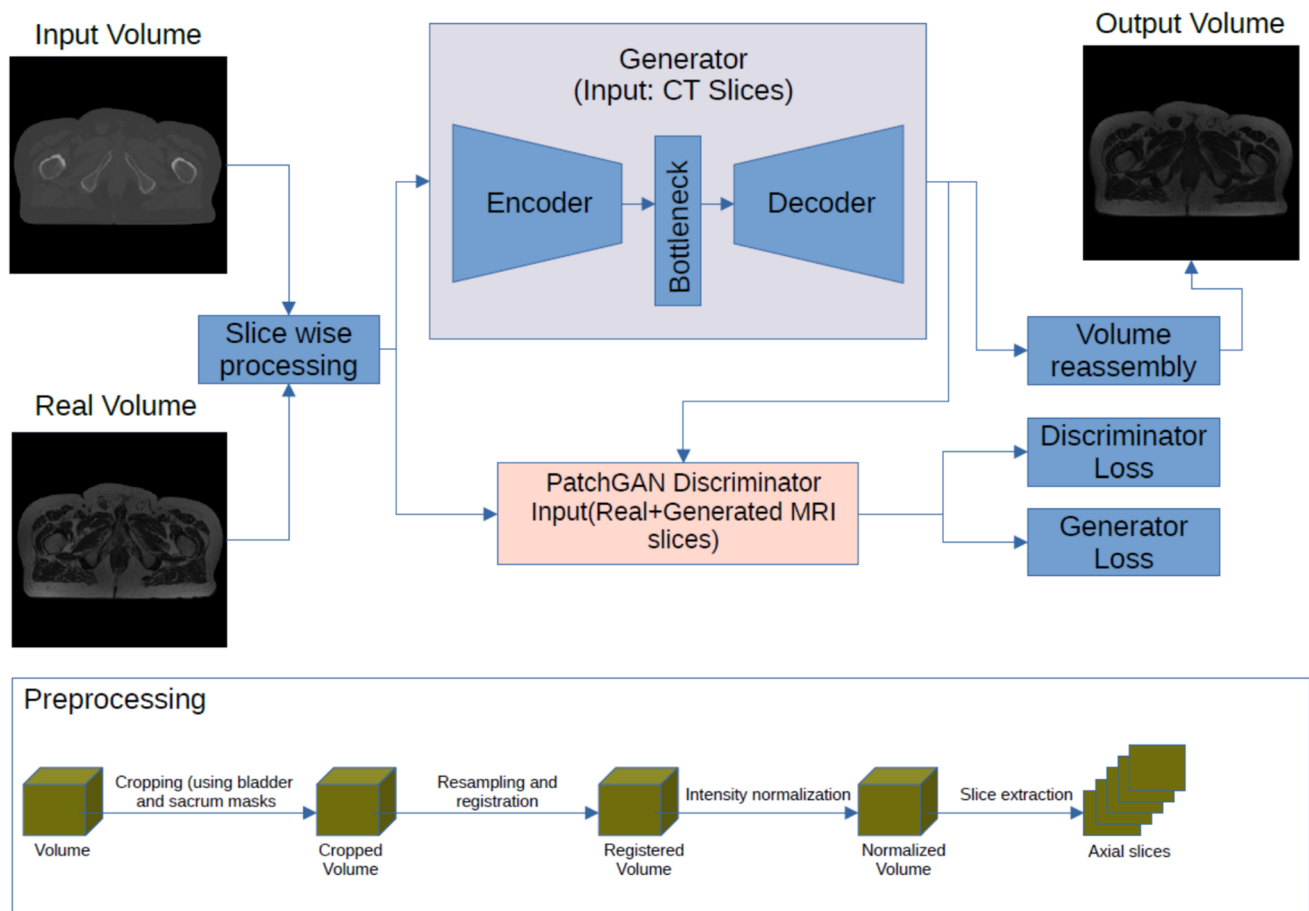


Fig. 1 Overview of the conditional GAN framework for Synthetic Pelvic MRI Generation from CT. The schematic illustrates the workflow for generating synthetic T2-weighted pelvic MRI from CT scans using a deep learning model. Top Section—**Input Volume**: The CT volume is pre-processed and converted to 2D axial slices via slice-wise processing. **Generator Network**: Each CT slice is processed by the generator (ESAU-Net, ResViT, or Cascade Gaze), comprising an encoder, bottleneck, and decoder, to synthesize corresponding MRI slices. **PatchGAN Discriminator**: Real and generated MRI slices are evaluated by a PatchGAN discriminator to compute adversarial losses (generator and discriminator loss) for improved realism. **Out-**

put Volume: Synthetic MRI slices are reassembled into a volumetric output. Bottom Section (Preprocessing)—**Volume Cropping**: Initial CT volumes are cropped using bladder and sacrum masks for anatomical centering. **Resampling & Registration**: Cropped volumes are resampled and registered to a standard anatomical space. **Intensity Normalization**: Volumes undergo intensity normalization to harmonize signal characteristics. **Slice Extraction**: Normalized volumes are decomposed into 2D axial slices for input to the deep learning model. Abbreviations: CT, computed tomography; MRI, magnetic resonance imaging; GAN, generative adversarial network

Generator Architectures

To identify the optimal architecture for CT-to-MRI synthesis, we conducted a comparative analysis of three state-of-the-art deep learning paradigms: a U-Net based architecture Efficient Self Attention U-Net (ESAU-Net) [11], a hybrid CNN-Transformer model (ResViT) [12], and a fully convolutional encoder-decoder based Cascaded Gaze Network (CG-Net) [13]. The goal was to evaluate which approach best balanced feature extraction, contextual understanding, and high-fidelity image generation for this specific task.

- **ESAU-Net (Efficient Self-Attention U-Net)**: This model enhances the classic U-Net framework by integrating a lightweight, channel-wise self-attention mechanism (Fig. 2) [11]. This allows the network to model long-range dependencies and compute contextually weighted feature maps with high computational efficiency, making it ideal for capturing both local details and global anatomical consistency.
- **ResViT (Residual Vision Transformer)**: This hybrid architecture combines a convolutional encoder for local feature extraction with a bottleneck of aggregated residual transformer blocks to learn global contextual

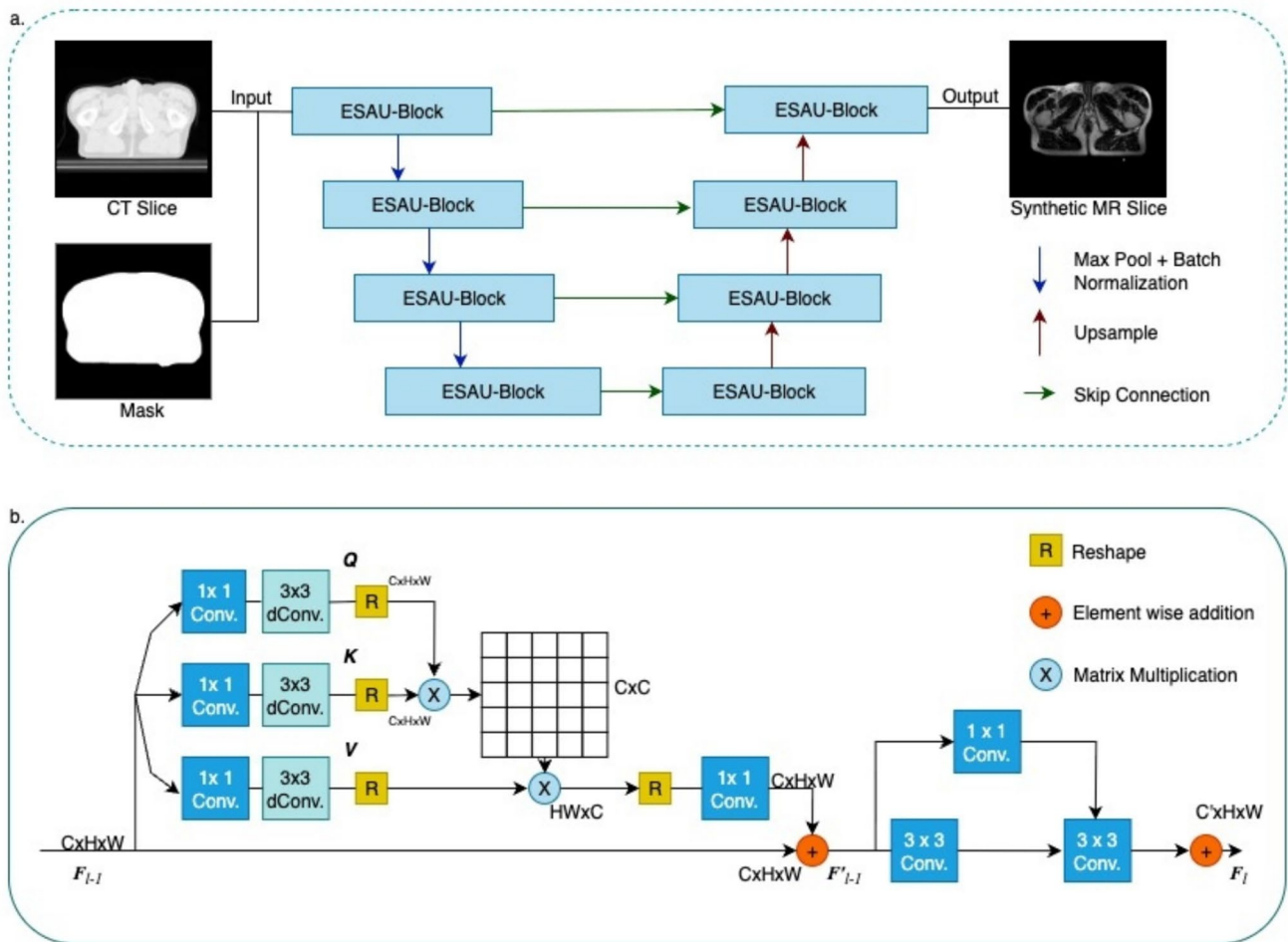


Fig. 2 Schematic of ESAUNet architecture and efficient self-attention block. **a** ESAUNet Model Architecture for Synthetic MRI Generation: This panel illustrates the workflow of the ESAUNet-based generator used for converting CT slices to synthetic MR slices. The input comprises a CT slice and its corresponding mask. The architecture consists of stacked ESAU-Blocks with skip connections, max pooling with batch normalization (downward blue arrows), and upsampling (upward red arrows). Skip connections (green arrows) transfer feature maps between encoding and decoding paths, facilitating the preservation of anatomical details. The output is a synthetic MR slice. **b** Detailed Structure of the ESAU-Block: This panel shows

the internal operation of the ESAU-Block, which enhances feature representation and long-range contextual modeling. Feature maps undergo 1×1 convolution followed by 3×3 depthwise convolution to generate query (Q), key (K), and value (V) embeddings. These are reshaped (yellow boxes, R), and matrix multiplication (blue circles, X) is used for attention computation. The attended features are further processed by convolutional layers and combined via element-wise addition (orange circles, +) to produce the final output feature map. Abbreviations: Conv, convolution; dConv, Depthwise Convolution; Q, Query; K, Key; V, Value; R, Reshape; X, matrix multiplication; +, element-wise addition

features [12]. By sharing parameters across its Transformer modules, it aims to prevent overfitting while capturing complex relationships in the data.

- **CG-Net (Cascaded gaze):** The CG-Net is an image restoration model based on a UNet encoder-decoder architecture, featuring a novel global context extractor (GCE) module [14]. The GCE efficiently captures long-range dependencies and multi-scale context through cascaded depthwise convolutions with expanding receptive fields and reduced spatial dimensions.

The detailed architectures are discussed in the supplementary material.

Model Training and Evaluation

Training Protocol

All models were trained on 256×256 2D axial slices for 100 epochs using the Adam optimizer (initial learning rate of 1×10^{-3}). A StepLR scheduler was employed to

gradually decay the learning rate by a factor of 0.5 every 10 epochs, promoting stable convergence.

Loss Function

We utilized a combination of pixel-based L1, perceptual, and adversarial loss functions based on ablation experiments (Table S1 and supplementary material) to optimize image fidelity:

- L1 Loss: Reduces pixel-level discrepancies.
- Perceptual Loss: Enforces high-level feature similarity using VGG19.
- Adversarial Loss: Motivates local texture synthesis via PatchGAN.

Our final loss is a weighted sum of the above losses.

Evaluation and Statistical Analysis

- Quantitative Metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Mean Squared Error (MSE), and Mean Absolute Error (MAE) were calculated per slice and averaged at patient and cohort levels (Table S2, supplementary material). To assess significant differences across the three model architectures, repeated measures one-way ANOVA was performed for each metric, with model as the within-subject factor ($k=3$) across the five cross-validation folds and five trained models evaluated on the independent test set. Given the small sample size ($n=5$), Friedman's non-parametric test was additionally performed. Post hoc pairwise comparisons were conducted using Bonferroni-corrected paired t -tests (adjusted significance threshold $\alpha=0.0167$). Effect sizes were reported as partial η^2 for ANOVA and Cohen's d for pairwise comparisons.
- Qualitative Assessment: Two abdominal radiologists (with 5.5 years of post-training experience, not involved in conceiving, designing, and conducting the study) performed blinded, crossover evaluation on synthetic and true MRI for the test cohort, rating overall image quality and anatomical delineation (rectal wall, sphincter, bladder, musculoskeletal, vascular structures) using five-point Likert scales (Table S3). The reading was done in two sessions, two weeks apart. Each session comprised a mix of true and synthetic MRI that were presented so that after both sessions, all true and synthetic scans had been read by the radiologists.
- Statistical Comparison: Mann–Whitney U test was used for qualitative scores; inter- and intra-observer agreement was assessed by weighted kappa statistics. Significance threshold was set at $p < 0.05$. Analyses were performed using Python (pandas, scipy, statsmodels).

The codebase is available at https://github.com/pks716/Translation_Model.

Results

Cross-Validation Performance

On the five-fold validation, all models showed competitive performance (Table 2). ResViT, with its transformer-based design, held a marginal lead in mean PSNR and SSIM over ESAU-Net and CG-Net in internal validation. However, parameter count did not translate into superior generalization.

Test Performance

ESAU-Net emerged as the optimal architecture on the Gold Atlas test set, achieving:

- PSNR: 22.21 dB (ResViT: 21.18, CascadedGazeNet: 21.41)
- SSIM: 0.748 (ResViT: 0.727, CascadedGazeNet: 0.703)
- MAE: 0.044 (lowest among tested models)

Table 3 shows the comparative performance of the different architectures on held-out test set.

Repeated measures ANOVA revealed significant main effects of model architecture on PSNR ($F(2,8)=9.43$, $p=0.008$, partial $\eta^2=0.70$) and SSIM ($F(2,8)=41.52$, $p<0.001$, partial $\eta^2=0.91$) for cross-validation (Table 1). Post hoc Bonferroni-corrected tests showed that CascadedGazeNet yielded significantly lower SSIM than both ESAU-Net ($p_{\text{adj}}=0.008$, $d=2.96$) and ResViT ($p_{\text{adj}}=0.002$, $d=4.30$), while ESAU-Net and ResViT did not differ significantly ($p_{\text{adj}}=1.00$). MSE and MAE did not reach significance after correction. On the independent test set (Table 2), ESAU-Net demonstrated significantly higher PSNR than ResViT ($p_{\text{adj}}=0.007$, $d=3.14$) and CascadedGazeNet ($p_{\text{adj}}=0.007$, $d=3.05$), and significantly lower MAE than both ($p_{\text{adj}}=0.009$ and 0.029 , respectively).

Qualitative Assessment by Radiologists

Table 4 and Figs. 3 and 4 show the comparison of image quality of true vs. sMRI.

- *General Features*: No statistically significant difference in perceived overall pelvic MRI quality ($p=0.294$) or

Table 2 fivefold cross-validation performance on the in-house validation set ($n = 18$ per fold)

Split/metric	PSNR			SSIM			MSE			MAE		
	ESANet	ResViT	Cascad- edGazeNet	ESANet	ResViT	Cascad- edGazeNet	ESANet	ResViT	CascadedGazeNet	ESANet	ResViT	Cascad- edGazeNet
Split 1	20.39	20.44	20.02	0.754	0.753	0.734	0.010	0.010	0.011	0.053	0.052	0.056
Split 2	24.27	24.49	24.24	0.776	0.782	0.769	0.004	0.004	0.004	0.028	0.028	0.028
Split 3	24.90	24.74	24.39	0.786	0.783	0.772	0.003	0.003	0.004	0.026	0.026	0.028
Split 4	23.95	23.76	23.36	0.762	0.762	0.747	0.004	0.005	0.005	0.031	0.032	0.033
Split 5	21.64	21.87	21.59	0.781	0.787	0.768	0.008	0.007	0.008	0.045	0.042	0.045
Mean	23.03	23.06	22.72	0.772	0.773	0.758	0.0058	0.0058	0.0064	0.037	0.036	0.038
RM-ANOVA	$F(2,8)=9.43, p=0.008^{\dagger}, \eta^2=0.70$			$F(2,8)=41.52, p<0.001^{\dagger}, \eta^2=0.91$			$F(2,8)=3.27, p=0.092, \eta^2=0.45$			$F(2,8)=4.79, p=0.043^{\dagger}, \eta^2=0.55$		
Post hoc	E vs R: 1.000			E vs R: 1.000			E vs R: 1.000			E vs R: 1.000		
(p_{adj})	E vs C: 0.166			E vs C: 0.008 [†]			E vs C: 0.212			E vs C: 0.240		
	R vs C: 0.002 [†]			R vs C: 0.002 [†]			R vs C: 0.212			R vs C: 0.142		

E ESANet, R ResViT, C CascadedGazeNet, RM repeated measures, F F statistics, η^2 partial eta squared. [†]Statistically significant. Post hoc:Bonferroni-corrected paired t -tests (significance threshold $p_{\text{adj}} < 0.05$)

Bold values refer to significant p -values (< 0.05)

Table 3 Performance on the independent test set ($n = 19$)

Model/metric	PSNR			SSIM			MSE			MAE		
	ESANet	ResViT	Cascaded-GazeNet	ESANet	ResViT	Cascaded-GazeNet	ESANet	ResViT	Cascaded-GazeNet	ESANet	ResViT	Cascaded-GazeNet
Model 1	21.18	19.93	20.69	0.686	0.652	0.681	0.012	0.014	0.012	0.053	0.059	0.054
Model 2	22.69	21.67	21.69	0.774	0.753	0.729	0.006	0.008	0.008	0.041	0.047	0.047
Model 3	22.52	21.14	21.40	0.753	0.746	0.707	0.008	0.008	0.010	0.043	0.048	0.049
Model 4	22.59	21.60	21.99	0.771	0.753	0.712	0.006	0.007	0.009	0.041	0.047	0.047
Model 5	22.07	21.55	21.26	0.756	0.729	0.687	0.008	0.008	0.011	0.044	0.046	0.052
Mean	22.21	21.18	21.41	0.748	0.727	0.703	0.008	0.009	0.010	0.044	0.049	0.050
RM-ANOVA	$F(2,8)=26.46, p<0.001^{\dagger}, \eta^2=0.87$			$F(2,8)=9.37, p=0.008^{\dagger}, \eta^2=0.70$			$F(2,8)=4.62, p=0.047^{\dagger}, \eta^2=0.54$			$F(2,8)=10.82, p=0.005^{\dagger}, \eta^2=0.73$		
Post-hoc (<i>p</i> -adj)	E vs R: 0.007 [†]			E vs R: 0.027 [†]			E vs R: 0.267			E vs R: 0.009 [†]		
	E vs C: 0.007 [†]			E vs C: 0.044 [†]			E vs C: 0.065			E vs C: 0.029 [†]		
	R vs C: 0.797			R vs C: 0.474			R vs C: 0.979			R vs C: 1.000		

E ESANet, R ResViT, C CascadedGazeNet, RM repeated measures, F F statistics, η^2 partial eta squared. [†]Statistically significant. Post hoc:Bonferroni-corrected paired t -tests (significance threshold $p_{\text{adj}} < 0.05$)

Bold values refer to significant p -values (< 0.05)

Table 4 Radiologist scoring of image quality on the test set

Category	Attribute	True MRI	Generated MRI	<i>p</i> -value
		Median (IQR) [range]	Median (IQR) [range]	
Rectal	Wall Delineation	4.0 (4.0–4.0) [3–5]	4.0 (4.0–4.0) [3–4]	0.358
	Sphincter Muscle Delineation	4.0 (4.0–5.0) [4–5]	4.0 (3.0–4.0) [3–4]	0.001†
	Motion Artefacts	4.0 (4.0–5.0) [3–5]	4.0 (4.0–4.0) [3–4]	0.054
	Overall Image Quality	4.0 (4.0–4.0) [3–5]	4.0 (3.0–4.0) [3–5]	0.230
Bladder	Wall Delineation	5.0 (5.0–5.0) [4–5]	4.0 (3.0–4.0) [2–5]	<0.001†
	VUJ Delineation	5.0 (4.0–5.0) [2–5]	4.0 (3.0–4.0) [2–5]	0.030†
	Artefacts	5.0 (5.0–5.0) [3–5]	4.0 (4.0–4.0) [3–5]	<0.001†
	Overall Image Quality	5.0 (4.0–5.0) [3–5]	4.0 (3.0–4.0) [3–4]	<0.001†
Musculoskeletal	Bone Cortex/Medulla Delineation	4.0 (4.0–4.0) [4–4]	4.0 (4.0–4.0) [3–4]	0.345
	Neural Foramina Delineation	4.0 (4.0–4.0) [4–5]	4.0 (4.0–4.0) [3–5]	0.099
	Joint Space/Cartilage Delineation	4.0 (4.0–4.0) [3–4]	4.0 (3.0–4.0) [3–5]	0.441
	Skeletal Muscle Delineation	5.0 (5.0–5.0) [4–5]	5.0 (4.0–5.0) [4–5]	0.091
	Neural Bundle Delineation	4.0 (3.0–4.0) [3–5]	4.0 (3.0–4.0) [2–5]	0.258
	Overall Image Quality	4.0 (4.0–4.0) [4–4]	4.0 (3.0–4.0) [3–4]	0.082
Vascular	Blood Vessel Wall Delineation	4.0 (4.0–4.0) [3–4]	4.0 (3.0–4.0) [3–4]	0.029†
Overall	Overall Pelvic MRI Image Quality	4.0 (4.0–4.0) [3–4]	4.0 (3.0–4.0) [3–4]	0.294

IQR, interquartile range (25th–75th percentile). †Statistically significant ($p < 0.05$, Mann–Whitney U test). Likert scale: 1=poor, 5=excellent

rectal wall delineation ($p = 0.358$) between synthetic and true MRI. Skeletal muscle and neural foramen delineation also showed non-significant differences.

- *Fine Details*: True MRI significantly outperformed synthetic MRI for sphincter muscle ($p = 0.001$), bladder wall ($p < 0.001$), and certain artifact ratings.
- *Artifacts and Failure Modes*: Synthetic MRI was more susceptible to distortion around fiducial markers and bladder contrast, as illustrated in Fig. 4.

Observer Agreement

Inter- and intra-observer agreement for anatomical ratings was robust, consistently rated “Good” to “Excellent” (weighted kappa 0.61–1.00, Table 5).

Discussion

Our study demonstrates the technical feasibility of generating synthetic T2w pelvic MRI from CT using advanced deep learning architectures, with ESAU-Net outperforming other approaches in both quantitative fidelity and practical image quality. The motivation to democratize access to MRI-equivalent anatomical imaging for pelvic cancer patients was validated by radiologist review, which found synthetic MRI statistically indistinguishable from true MRI for general anatomical assessment and rectal wall delineation. This

supports the potential of synthetic MRI to augment radiotherapy planning and computational research in settings where MRI access is limited.

Comparing our quantitative metrics to previous literature, ESAU-Net achieved a mean PSNR of 22.21 dB and SSIM of 0.748 on the independent test set. Kalantar et al. in one of the few studies addressing pelvic CT-to-MRI synthesis reported a PSNR of 18.28 for a CycleGAN model and 20.17 for a standard UNet, both on T1-weighted images [9]. Our results substantially exceed these benchmarks for T2w synthesis, reflecting the benefit of advanced attention-based architectures and optimized perceptual loss functions. Ni et al. recently proposed a multi-dimensional diffusion architecture, achieving a PSNR of 16.65 dB on the SynthRAD2023 dataset [6]. ESAU-Net model achieved markedly higher fidelity. These comparisons underscore that efficient, context-aware models rather than sheer parameter magnitude or generative complexity are key for high-quality CT-to-MRI synthesis in the pelvis.

Our study has several limitations that warrant discussion. First, the sample size is limited ($n = 90$ for training/validation, $n = 19$ for independent testing), which constrains statistical power and generalizability. The small number of cross-validation folds ($n = 5$) limits the precision of performance estimates, and despite large effect sizes, some pairwise comparisons did not achieve statistical significance after correction. We position this work as a technical feasibility study; larger, prospective cohorts are needed to confirm these findings.

Second, the multi-center, multi-scanner dataset, while enhancing external validity, introduces significant

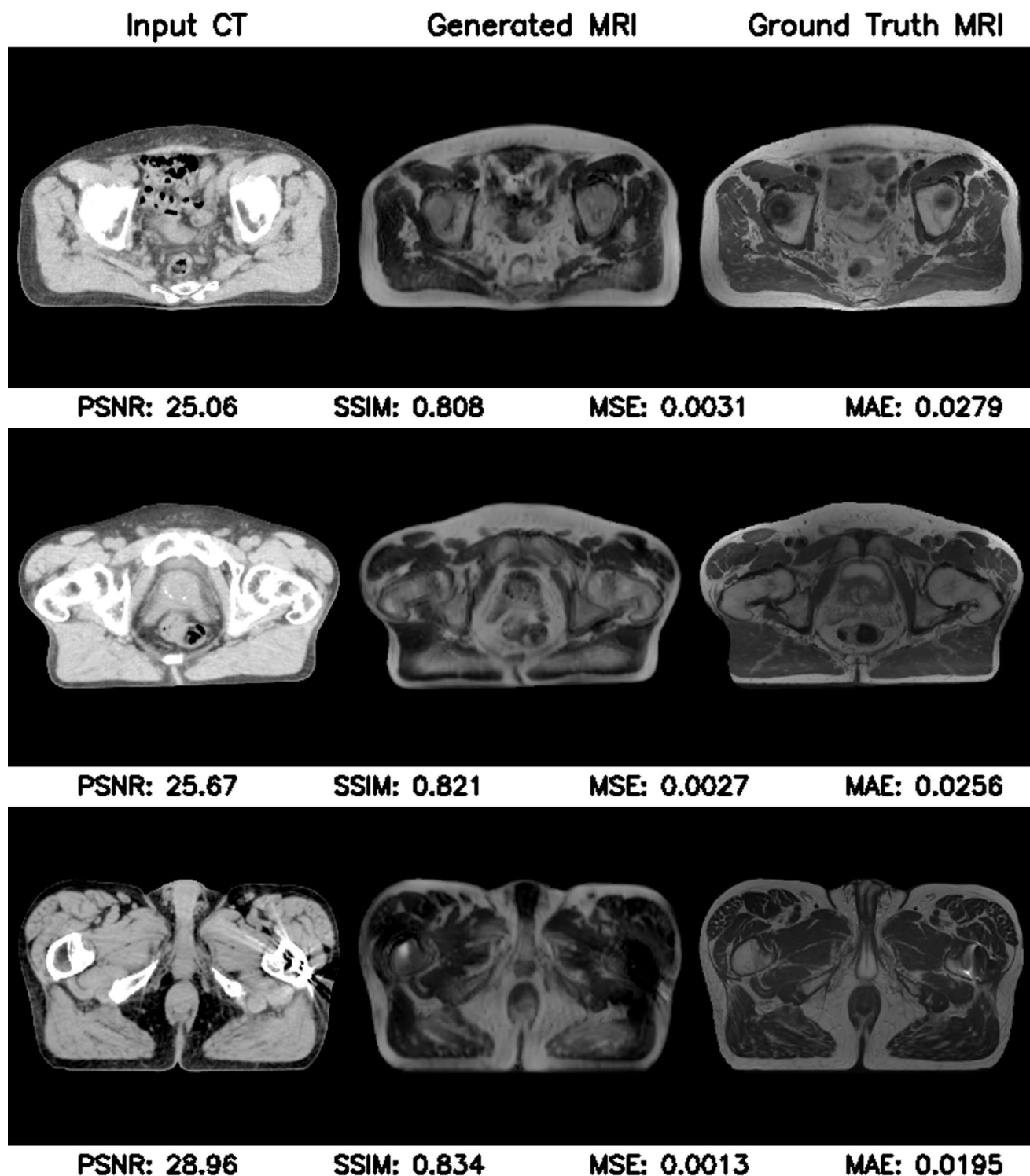
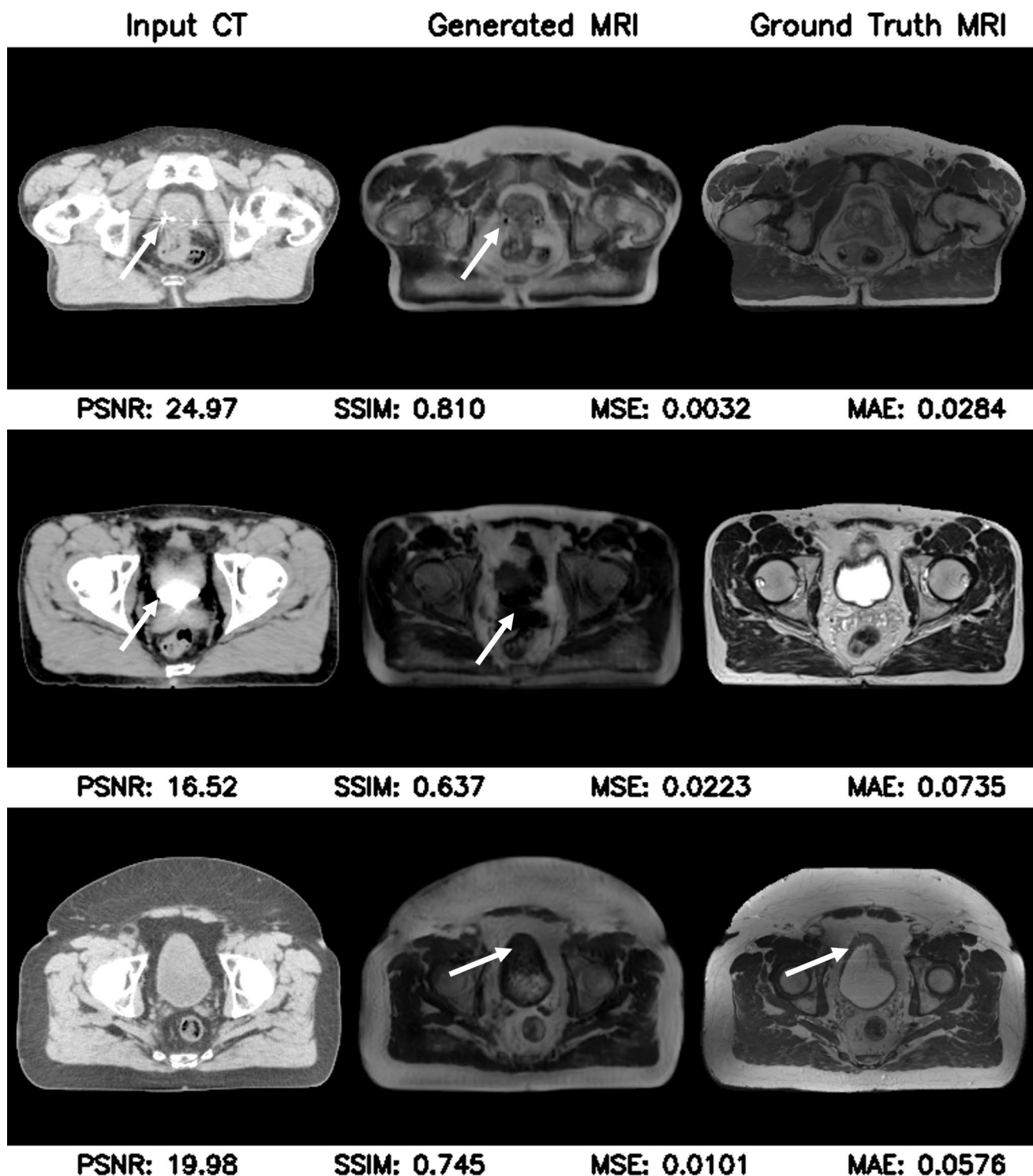


Fig. 3 Visual Comparison of input CT, generated MRI, and ground-truth MRI slices. Representative axial pelvic slices are shown for three example cases. Left column: input CT image. Middle column: high-quality synthetic MRI generated from the corresponding CT using the proposed deep learning framework. Right column: ground-truth T2-weighted MRI. Quantitative metrics (PSNR, SSIM, MSE, MAE) for each row demonstrate the fidelity of the generated MRI com-

pared to the ground-truth. The synthetic MRI closely matches the anatomical detail and soft-tissue contrast of the real MRI, supporting the effectiveness of the proposed model for high-quality pelvic MRI synthesis. Abbreviations: CT, computed tomography; MRI, magnetic resonance imaging; PSNR, peak signal-to-noise ratio; SSIM, Structural Similarity Index; MSE, mean squared error; MAE, mean absolute error



heterogeneity in imaging protocols, scanner manufacturers, and field strengths (1.5 T vs 3 T). This diversity complicates universal algorithm development and limits homogeneous validation. The reliance on retrospective data with incomplete patient metadata and privacy constraints restricts subgroup analysis (e.g., by age, sex, body habitus, disease status) and

precludes comprehensive assessment of algorithmic bias across demographic groups.

Third, potential selection bias exists: only patients with paired CT and MRI were included, which may not represent the broader clinical population requiring synthetic MRI. The exclusion of deformed CTs from training, though

Fig. 4 Examples of artifacts and poor quality in generated synthetic MRI. Representative axial pelvic slices are shown for three cases, illustrating common failure modes in synthetic MRI generation. Top row: The presence of a fiducial marker in the input CT (arrow) leads to significant distortion of the prostate anatomy in the generated MRI (arrow), which is not present in the ground-truth MRI. Middle row: Dense contrast material within the bladder in the CT scan (arrow) results in severe distortion of the urinary bladder and adjacent structures in the generated MRI (arrow) compared to the true MRI. Bottom row: An artifact is observed in the anterior bladder wall; thickening is evident in the ground-truth MRI but not on CT, and the generated MRI shows a pronounced artifact at the site of thickening (arrow). For each case, quantitative metrics (PSNR, SSIM, MSE, MAE) are provided, highlighting the degradation in image fidelity where artifacts are present. These examples demonstrate the limitations of synthetic MRI in cases where CT contains significant artifacts or anatomical ambiguities. Abbreviations: CT =, computed tomography; MRI =, magnetic resonance imaging; PSNR =, peak signal-to-noise ratio; SSIM =, Structural Similarity Index; MSE =, mean squared error; MAE =, mean absolute error

necessary for fair benchmarking, further restricts paired learning and may limit applicability to post-surgical or pathologically distorted anatomy.

Fourth, preprocessing steps such as resampling and intensity normalization, while necessary for harmonization, can degrade anatomical fidelity by smoothing critical structures or introducing interpolation artifacts. Our models were trained on 2D slices to manage computational resources, which may compromise volumetric continuity and inter-slice context—a limitation that 2.5D or 3D architectures could address in future work.

Fifth, limitations persist in fine structure reconstruction: synthetic MRI remains inferior to true MRI for high-detail anatomical features such as sphincter muscle and bladder wall delineation, as confirmed by significantly lower radiologist scores (Table 3). Generated images are also more prone to artifacts from fiducial markers or residual contrast. These deficits reflect the inherent challenge of inferring rich MRI soft-tissue contrast from CT's limited contrast resolution.

Finally, qualitative assessment was performed by only two radiologists from a single institution, which may limit generalizability of perceptual findings. Although inter- and intra-observer agreement was strong, evaluation by a larger, multi-institutional panel would strengthen clinical validation.

Despite these limitations, this study establishes a robust technical and clinical benchmark for pelvic CT-to-MRI synthesis. The superior performance of ESAUNet, validated across quantitative metrics and qualitative radiologist assessment, supports the broader adoption of efficient attention-based architectures for cross-modality image translation.

In conclusion, conditional GAN frameworks leveraging attention-based architecture offer a promising solution for CT-to-T2-weighted pelvic MRI translation. ESAUNet's efficiency and robust performance establish a foundation for advancing cross-modality synthesis, addressing MRI access disparities, and unlocking new computational applications in resource-limited settings.

Table 5 Detailed observer agreement for true vs. generated MRI (weighted Kappa, κ)

Category	Attribute	Inter-observer agreement	Intra-observer agreement R1	Intra-observer agreement R2
		True/Generated	True/Generated	True/Generated
Rectal	Wall Delineation	0.88/0.75	0.91/0.87	0.90/0.86
	Sphincter Delineation	0.86/0.68	0.89/0.85	0.88/0.83
	Motion Artefacts	0.84/0.71	0.87/0.83	0.85/0.82
	Overall Image Quality	0.87/0.78	0.90/0.86	0.89/0.85
Bladder	Wall Delineation	0.94/0.72	0.95/0.92	0.93/0.90
	VUJ Delineation	0.85/0.65	0.88/0.83	0.86/0.82
	Artefacts	0.93/0.74	0.94/0.90	0.92/0.89
	Overall Image Quality	0.91/0.76	0.93/0.90	0.92/0.89
Musculoskeletal	Bone Cortex/Medulla	0.92/0.78	0.94/0.91	0.93/0.92
	Neural Foramina	0.88/0.61	0.90/0.87	0.89/0.86
	Joint Space/Cartilage	0.85/0.64	0.88/0.84	0.87/0.83
	Skeletal Muscle	0.92/0.77	0.93/0.91	0.94/0.92
	Neural Bundle	0.83/0.55	0.86/0.82	0.85/0.81
Vascular	Overall Image Quality	0.90/0.75	0.92/0.90	0.91/0.89
	Blood Vessel Wall	0.84/0.58	0.87/0.83	0.86/0.82
Overall	Overall MRI Quality	0.90/0.76	0.92/0.89	0.91/0.88

R1: reader 1 and R2: reader 2. Interpretation of Kappa Value: <0.00= Poor; 0.00–0.20= Slight; 0.21–0.40= Fair; 0.41–0.60= Moderate; 0.61–0.80= Substantial; 0.81–1.00= Almost perfect

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Data Availability Data related to the manuscript will be shared upon reasonable request as per the institutional policy.

Declarations

Ethics Approval The study was approved by the Institutional Ethics Committee (PGI/IEC/2025/EICOO0572).

Consent to Participate Informed consent was obtained from all individual participants included in the study.

Consent for Publication Not required as no patient identifying information is shown.

Competing interests The authors declare no competing interests.

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